

**Nick's Question 0.1** In the solution (1.8)

$$u(t, x) = J_0(t, x) + I(t, x)$$

Why isn't the  $I_t^\gamma$  operator represented? It seems like (1.8) is the same whether  $I_t^\gamma$  is there or not.

**Solution:** It is built into the triplet in equation (1.5)

**Nick's Question 0.2 Proof of Theorem 3.3**

Why is (3.5) due to (3.1) and (3.3). I see why (3.1) is important but why (3.3)? Does  $\bar{\mathcal{K}}$  satisfy the sup property? This may be Theorem 3.4?

**Nick's Question 0.3 Theorem 3.5** What happens to the  $\hat{C}_t$  when we are finding the upper bound for  $\bar{m}_p$ . It seems like the

$$\frac{1}{t} \log \hat{C}_t$$

terms are being absorbed into the constant as  $t \rightarrow \infty$ . Is this true because of

$$\frac{1}{t} \log(t) \rightarrow 0 \quad t \rightarrow \infty?$$

**Nick's Question 0.4 possible typo on page 5087 equation 4.16 -4.18** I think that in the asymptotic bound for  $Y$ , that  $\gamma$  needs to be replaced with  $1 - \beta$  not just 1 for the  $Z^*$  case. This can be seen from (4.4)-(4.6)

**Nick's Question 0.5**

1. Where does the  $\int_0^t d\tau$  come from and why can we remove the laplace transform from  $f$  in the proof of theorem 4.1. I see using

$$\mathcal{L}[t^{\beta-1} E_{\alpha,\beta}(-\lambda t^\alpha)](s) = \frac{s^{\alpha-\beta}}{s^\alpha + \lambda}$$

then

$$\frac{s^{-\gamma}}{s^\beta + \frac{\nu}{2} |\xi|^\alpha} = \mathcal{L} \left[ t^{\beta+\gamma-1} E_{\beta,\beta+\gamma} \left( -\frac{\nu}{2} |\xi|^\alpha t^\beta \right) \right] (s)$$

so from the right term from the top of page 5090 we have

$$\mathcal{L} \left[ t^{\beta+\gamma-1} E_{\beta,\beta+\gamma} \left( -\frac{\nu}{2} |\xi|^\alpha t^\beta \right) \right] (s) \tilde{f}(s, \xi)$$

but I do not know why we can use this to get the  $\int_0^t d\tau$  term in the expression for  $\hat{u}(t, \xi)$ .

2. In order to show that  $Y(t, x) = {}_t D_+^\theta Z(t, x)$  you showed that

$$\mathcal{F}(Y(t, \cdot))(\xi) = {}_t D_+^\theta \mathcal{F}(Z(t, \cdot))(\xi)$$

Does the  ${}_tD_+^\theta$  operator commute with the Fourier transform?

### Nick's Question 0.6 Proof of Lemma 4.11

1. In the beginning of the proof it says that since  $\Delta$  and  $\delta$  are 0 and 1 respectively then Theorem 1.3 implies that for all  $0 < x < 1$  that... But this is not a Theorem from this paper. Which paper is this theorem found in?
2. Is (4.31) missing a  $1/\pi$ . Also should there be a  $\pi$  in the power of the  $e$  in the following line. We have the following formulas

- $\Gamma(1+z) = z\Gamma(z)$
- $\sin(\pi x) = \frac{\pi}{-x\Gamma(x)\Gamma(-x)} \implies \sin(\pi k\theta) = \frac{\pi}{-k\theta\Gamma(k\theta)\Gamma(-k\theta)}$

and using these we see that

$$\begin{aligned} \sum_k \frac{(-1)^k}{k!} \frac{\Gamma(k+1)}{\Gamma(-k\theta)\Gamma(1+k\theta)} x^{k/\theta} &= \sum_k \frac{-(-1)^k}{\pi} \frac{\pi}{-k\theta\Gamma(-k\theta)\Gamma(k\theta)} x^{k/\theta} \\ &= \sum_k \frac{-(-1)^k}{\pi} \sin(\pi k\theta) x^{k/\theta} \\ &= -\frac{1}{\pi} \text{IM} \left( \sum_k (-1)^k e^{\pi k\theta i} x^{k/\theta} \right) \\ &= -\frac{1}{\pi} \text{IM} \frac{1}{1 + e^{\pi\theta i} x^{1/\theta}} \end{aligned}$$

I am not sure if there is a easy way to find the imaginary part and my wolfram subscription ran out so I can't use that. The same problem happens with the next group of centered equations on the same page.

### Nick's Question 0.7 Proof lemma 4.12

1. There is a small typo, right above where it says part(b), it should be  $H_{1,2}^{2,0}$ . Details for part a. We want to show that

$$H_{2,3}^{2,1} \left( x^2 \middle| \begin{matrix} (d, 2) & (\mu, \theta) \\ (d/2, 1) & (1, 1) \end{matrix} \begin{matrix} (1+d, 2) \end{matrix} \right) = -2H_{1,2}^{2,0} \left( x^2 \middle| \begin{matrix} (\mu, \theta) \\ ((d+1)/2, 1) \end{matrix} \begin{matrix} (1, 1) \end{matrix} \right)$$

We have for the left hand side, just considering the  $H_{23}^{21}$  part of  $H_{2,3}^{2,1}$

$$\begin{aligned} H_{23}^{21} \left( x^2 \middle| \begin{matrix} (d, 2) & (\mu, \theta) \\ (d/2, 1) & (1, 1) \end{matrix} \begin{matrix} (1+d, 2) \end{matrix} \right) &= \frac{\Gamma(\frac{d}{2} + s)\Gamma(1+s)\Gamma(1-d-2s)}{\Gamma(\mu + \theta s)\Gamma(1-1-d-2s)} \\ &= \frac{\Gamma(\frac{d}{2} + s)\Gamma(1-d-2s)}{\Gamma(-d-2s)} * \frac{\Gamma(1+s)}{\Gamma(\mu + \theta s)} \\ &= -2 \frac{\Gamma(1+d/2+s)\Gamma(1+s)}{\Gamma(\mu + \theta s)} \\ &= -2H_{12}^{20} \left( x^2 \middle| \begin{matrix} (\mu, \theta) \\ ((d+2)/2, 1) \end{matrix} \begin{matrix} (1, 1) \end{matrix} \right) \end{aligned}$$

2. There is a typo in part (b), from the definition of  $f$  and  $H$  we need

$$\begin{aligned} f_{d+1,\mu,\theta}(x) &= x^{-(d+1)} H_{1,2}^{2,0} \left( x^2 \left| \begin{matrix} (\mu, \theta) \\ ((d+1)/2, 1) & (1, 1) \end{matrix} \right. \right) \\ &= x^{-(d+1)} \frac{1}{2\pi i} \int_{Li\gamma\infty} ds \frac{\Gamma((d+1)/2 + s) \Gamma(1 + s)}{\Gamma(\mu + \theta s)} \end{aligned}$$

but this does not change anything since all the integral manipulations had to do with  $x$  not  $s$ .

### Nick's Question 0.8 Proof of Theorem 4.6

1. Why are we considering

$$g(x) = x^{-d} H_{2,3}^{2,1} \left( x^\alpha \left| \begin{matrix} (1, 1) & (\beta, \beta) \\ (d/2, \alpha/2) & (1, 1) & (1, \alpha/2) \end{matrix} \right. \right)?$$

In equation (4.6)  $Y$  with  $\gamma = 0$  and  $t = 1$  is given by

$$Y \propto |x|^{-d} H_{2,3}^{2,1} \left( x^\alpha \left| \begin{matrix} (1, 1) & (\beta, \beta) \\ (d/2, \alpha/2) & (1, 1) & (1, \alpha/2) \end{matrix} \right. \right)$$

So why do we have the  $x^{-d}$  term in  $g(x)$  above instead of  $|x|^{-d}$  term?

2. Also there is a small typo when citing Lemma 4.3 in the conclusion of part (3) of the proof. You said it follows by case 6 but you meant to say case 5.